

For bipartite graphs H , Corollary 7.1.3 says that substantially fewer than $\binom{n}{2}$ edges suffice to force an H subgraph. It turns out that

$$c_1 n^{2 - \frac{2}{r+1}} \leq \text{ex}(n, K_{r,r}) \leq c_2 n^{2 - \frac{1}{r}}$$

for suitable constants c_1, c_2 depending on r ; the lower bound is obtained by random graphs,² the upper bound is calculated in Exercise 12. If H is a forest, then $H \subseteq G$ as soon as $\varepsilon(G)$ is large enough, so $\text{ex}(n, H)$ is at most linear in n (Exercise 14). Erdős and Sós conjectured in 1963 that $\text{ex}(n, T) \leq \frac{1}{2}(k-1)n$ for all trees with $k \geq 2$ edges; as a general bound for all n , this is best possible for every T (Exercises 15–17).

*Erdős-Sós
conjecture*

A related but rather different question is whether large values of ε or χ can force a graph G to contain a given tree T as an *induced* subgraph. Of course, we need some additional assumption for this to make sense – for example, to prevent G from just being a large complete graph. The weakest sensible such assumption is that G has bounded clique number, i.e., that $G \not\supseteq K^r$ for some fixed integer r . Then large average degree still does not force an induced copy of T – consider complete bipartite graphs – but large chromatic number might: according to the Gyárfás-Sumner conjecture from Chapter 5.6, there exists for every $r \in \mathbb{N}$ and every tree T an integer $k = k(T, r)$ such that every graph G with $\omega(G) \leq r$ and $\chi(G) > k$ contains T as an induced subgraph.

7.2 Minors

In this section and the next we ask to what extent assumptions about invariants of a graph such as average degree, chromatic number, or girth can force it to contain another given graph as a minor or topological minor.

As a starting question, let us consider the analogue of Turán's theorem: how many edges on n vertices force a K^r minor or topological minor? The qualitative answer is that, unlike for K^r subgraphs where we might need as many as $\frac{1}{2} \frac{r-2}{r-1} n^2$ edges, a number of edges linear in n is enough: it suffices to assume that the graph has large enough average degree (depending on r).

Proposition 7.2.1. *Every graph of average degree at least 2^{r-2} has a K^r minor, for all $r \in \mathbb{N}$.*

Proof. We apply induction on r . For $r \leq 2$ the assertion is trivial. For the induction step let $r \geq 3$, and let G be any graph of average degree at least 2^{r-2} . Then $\varepsilon(G) \geq 2^{r-3}$; let H be a minimal minor of G with $\varepsilon(H) \geq 2^{r-3}$. Pick a vertex $x \in H$. By the minimality of H , x is not iso-

² see Chapter 11

lated. And each of its neighbours y has at least 2^{r-3} common neighbours with x : otherwise contracting the edge xy would lose us one vertex and at most 2^{r-3} edges, yielding a smaller minor H' with $\varepsilon(H') \geq 2^{r-3}$. The subgraph induced in H by the neighbours of x therefore has minimum degree at least 2^{r-3} , and hence has a K^{r-1} minor by the induction hypothesis. Together with x this yields the desired K^r minor of G . $\square$

In Proposition 7.2.1 we needed an average degree of 2^{r-2} to force a K^r minor by induction on r . Forcing a topological K^r minor is a little harder: we shall fix its branch vertices in advance and then construct its subdivided edges inductively, which requires an average degree of $2^{\binom{r}{2}}$ to start with. Apart from this difference, the proof follows the same idea:

Proposition 7.2.2. *Every graph of average degree at least $2^{\binom{r}{2}}$ has a topological K^r minor, for every integer $r \geq 2$.*

Proof. The assertion is clear for $r = 2$, so let us assume that $r \geq 3$. We show by induction on $m = r, \dots, \binom{r}{2}$ that every graph G of average degree $d(G) \geq 2^m$ has a topological minor X with r vertices and m edges.

If $m = r$ then, by Propositions 1.2.2 and 1.3.1, G contains a cycle of length at least $\varepsilon(G) + 1 \geq 2^{r-1} + 1 \geq r + 1$, and the assertion follows with $X = C^r$.

Now let $r < m \leq \binom{r}{2}$, and assume the assertion holds for smaller m . Let G with $d(G) \geq 2^m$ be given; thus, $\varepsilon(G) \geq 2^{m-1}$. Since G has a component C with $\varepsilon(C) \geq \varepsilon(G)$, we may assume that G is connected. Consider a maximal set $U \subseteq V(G)$ such that U is connected in G and $\varepsilon(G/U) \geq 2^{m-1}$; such a set U exists, because G itself has the form G/U with $|U| = 1$. Since G is connected, we have $N(U) \neq \emptyset$.

Let $H := G[N(U)]$. If H has a vertex v of degree $d_H(v) < 2^{m-1}$, we may add it to U and obtain a contradiction to the maximality of U : when we contract the edge vv_U in G/U , we lose one vertex and $d_H(v) + 1 \leq 2^{m-1}$ edges, so ε will still be at least 2^{m-1} . Therefore $d(H) \geq \delta(H) \geq 2^{m-1}$. By the induction hypothesis, H contains a TY with $|Y| = r$ and $\|Y\| = m - 1$. Let x, y be two branch vertices of this TY that are non-adjacent in Y . Since x and y lie in $N(U)$ and U is connected in G , G contains an x - y path whose inner vertices lie in U . Adding this path to the TY , we obtain the desired TX . $\square$

In Chapter 3.5 we used the TK^r from Proposition 7.2.2 (stated there as Lemma 3.5.1) for a first proof that large enough connectivity $f(k)$ implies that a graph is k -linked. Later, in Theorem 3.5.3, we saw that connectivity as low as $2k$, coupled with an average degree of at least $16k$, is enough to imply this.

Conversely, we can use the more involved Theorem 3.5.3 to reduce the bound in Proposition 7.2.2 from exponential to quadratic, which is best possible up to a multiplicative constant (Exercise 24):

Theorem 7.2.3. *There is a constant $c \in \mathbb{R}$ such that, for every $r \in \mathbb{N}$, every graph G of average degree $d(G) \geq cr^2$ contains K^r as a topological minor.*

Proof. We prove the theorem with $c = 10$. Let G with $d(G) \geq 10r^2$ be given. By Theorem 1.4.3 for $k := r^2$, G has a subgraph H with $\kappa(H) \geq r^2$ and $\varepsilon(H) > \varepsilon(G) - r^2 \geq 4r^2$. For a TK^r in H , pick a set X of r vertices in H as branch vertices, and a set Y of $r(r-1)$ neighbours of X in H , $r-1$ for each vertex in X , as initial subdividing vertices. These are r^2 vertices altogether; they can be chosen distinct, since $\delta(H) \geq \kappa(H) \geq r^2$.

It remains to link up the vertices of Y in pairs, by disjoint paths in $H' := H - X$ corresponding to the edges of K^r . This can be done if Y is linked in H' . We show more generally that H' is $\frac{1}{2}r(r-1)$ -linked, by checking that H' satisfies the premise of Theorem 3.5.3 for $k = \frac{1}{2}r(r-1)$. We have $\kappa(H') \geq \kappa(H) - r \geq r(r-1) = 2k$. And as H' was obtained from H by deleting at most $r|H|$ edges (as well as some vertices), we also have $\varepsilon(H') \geq \varepsilon(H) - r \geq 4r(r-1) = 8k$. $\square$

(1.4.3)
(3.5.3)

For small r one can try to determine the exact number of edges needed to force a TK^r subgraph on n vertices. For $r = 4$, this number is $2n - 2$; see Corollary 7.3.2. For $r = 5$, plane triangulations yield a lower bound of $3n - 5$ (Corollary 4.2.10). The converse, that $3n - 5$ edges do force a TK^5 – not just either a TK^5 or a $TK_{3,3}$, as they do by Corollary 4.2.10 and Kuratowski's theorem – is already a difficult theorem (Mader 1998).

The average degree needed to force an arbitrary K^r minor is less than that for a TK^r , and it is known very precisely; see the notes for the value of c in the following result.

Theorem 7.2.4. (Kostochka 1982)

There exists a constant $c \in \mathbb{R}$ such that, for every $r \in \mathbb{N}$, every graph G of average degree $d(G) \geq cr\sqrt{\log r}$ contains K^r as a minor. Up to the value of c , this bound is best possible as a function of r .

The easier implication of the theorem, the fact that in general an average degree of $cr\sqrt{\log r}$ is needed to force a K^r minor, follows from considering random graphs as introduced in Chapter 11. The converse implication, that this average degree suffices, is proved by methods not dissimilar to the proof of Theorem 3.5.3.

Rather than proving Theorem 7.2.4, therefore, we devote the remainder of this section to another striking aspect of forcing minors: that we can force a K^r minor in a graph simply by raising its girth (as long as we do not merely subdivide edges). At first glance, this may seem almost paradoxical. But it looks more plausible if, rather than trying to force a K^r minor directly, we instead try to force a minor just of large

minimum or average degree – which suffices by Theorem 7.2.4. For if the girth g of a graph is large then the ball $\{v \mid d(x, v) < \lfloor g/2 \rfloor\}$ around a vertex x induces a tree with many leaves, each of which sends all but one of its incident edges away from the tree. Contracting enough disjoint such trees we can thus hope to obtain a minor of large average degree, which in turn will have a large complete minor.

The following lemma realizes this idea.

Lemma 7.2.5. *Let $d, k \in \mathbb{N}$ with $d \geq 3$, and let G be a graph of minimum degree $\delta(G) \geq d$ and girth $g(G) \geq 8k + 3$. Then G has a minor H of minimum degree $\delta(H) \geq d(d-1)^k$.*

Proof. Let $X \subseteq V(G)$ be maximal with $d(x, y) > 2k$ for all distinct $x, y \in X$. For each $x \in X$ put $T_x^0 := \{x\}$. Given $i < 2k$, assume that we have defined disjoint trees $T_x^i \subseteq G$ (one for each $x \in X$) whose vertices together are precisely the vertices at distance at most i from X in G . Joining each vertex at distance $i+1$ from X to a neighbour at distance i , we obtain a similar set of disjoint trees T_x^{i+1} . As every vertex of G has distance at most $2k$ from X (by the maximality of X), the trees $T_x := T_x^{2k}$ obtained in this way partition the entire vertex set of G . Let H be the minor of G obtained by contracting every T_x .

To prove that $\delta(H) \geq d(d-1)^k$, note first that the T_x are induced subgraphs of G , because $\text{diam}(T_x) \leq 4k$ and $g(G) > 4k+1$. Similarly, there is at most one edge in G between any two trees T_x and T_y : two such edges, together with the paths joining their ends in T_x and T_y , would form a cycle of length at most $8k+2 < g(G)$. So all the edges leaving T_x are preserved in the contraction.

How many such edges are there? Note that, for every vertex $u \in T_x^{k-1}$, all its $d_G(u) \geq d$ neighbours v also lie in T_x : since $d(v, x) \leq k$ and $d(x, y) > 2k$ for every other $y \in X$, we have $d(v, y) > k \geq d(v, x)$, so v was added to T_x rather than to T_y when those trees were defined. Therefore T_x^k , and hence also T_x , has at least $d(d-1)^{k-1}$ leaves. But every leaf of T_x sends at least $d-1$ edges away from T_x , so T_x sends at least $d(d-1)^k$ edges to (distinct) other trees T_y . $\square$

Lemma 7.2.5 provides Theorem 7.2.4 with the following corollary:

Theorem 7.2.6. (Thomassen 1983)

There exists a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that every graph of minimum degree at least 3 and girth at least $f(r)$ has a K^r minor, for all $r \in \mathbb{N}$.

Proof. We prove the theorem with $f(r) := 8 \log r + 4 \log \log r + c$, for some constant $c \in \mathbb{R}$. Let $k = k(r) \in \mathbb{N}$ be minimal with $3 \cdot 2^k \geq c' r \sqrt{\log r}$, where $c' \in \mathbb{R}$ is the constant from Theorem 7.2.4. Then for a suitable constant $c \in \mathbb{R}$ we have $8k+3 \leq 8 \log r + 4 \log \log r + c$, and the result follows by Lemma 7.2.5 and Theorem 7.2.4. $\square$

Large girth can also be used to force a topological K^r minor. We now need some vertices of degree at least $r - 1$ to serve as branch vertices, but if we assume a minimum degree of $r - 1$ to secure these, we can even get by with a girth bound that is independent of r :

Theorem 7.2.7. (Kühn & Osthus 2002)

[7.3.9]

There exists a constant g such that $G \supseteq TK^r$ for every graph G satisfying $\delta(G) \geq r - 1$ and $g(G) \geq g$.

7.3 Hadwiger's conjecture

As we saw in Section 7.2, an average degree of $cr\sqrt{\log r}$ suffices to force an arbitrary graph to have a K^r minor, and an average degree of cr^2 forces it to have a topological K^r minor. If we replace 'average degree' above with 'chromatic number' then, with almost the same constants c , the two assertions remain true: this is because every graph with chromatic number k has a subgraph of average degree at least $k - 1$ (Lemma 5.2.3).

Although both functions above, $cr\sqrt{\log r}$ and cr^2 , are best possible (up to the constant c) for the said implications with 'average degree', the question arises whether they are still best possible with 'chromatic number' – or whether some slower-growing function would do in that case. What lies hidden behind this problem about growth rates is a fundamental question about the nature of the invariant χ : can this invariant have some direct *structural* effect on a graph in terms of forcing concrete substructures, or is its effect no greater than that of the 'unstructural' property of having lots of edges somewhere, which it implies trivially?

Neither for general nor for topological minors is the answer to this question known. For general minors, however, the following conjecture of Hadwiger suggests a positive answer:

Conjecture. (Hadwiger 1943)

The following implication holds for every integer $r > 0$ and every graph G :

$$\chi(G) \geq r \Rightarrow G \succcurlyeq K^r.$$

Hadwiger's conjecture is trivial for $r \leq 2$, easy for $r = 3$ and $r = 4$ (exercises), and equivalent to the four colour theorem for $r = 5$ and $r = 6$. For $r \geq 7$ the conjecture is open, but it is true for line graphs (Exercise 34) and for graphs of large girth (Exercise 32; see also Corollary 7.3.9). Rephrased as $G \succcurlyeq K^{\chi(G)}$, it is true for almost all graphs.³ In general, the conjecture for $r + 1$ implies it for r (exercise).

³ See Chapter 11 for the notion of 'almost all'.